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A STRUCTURAL AND DEPOSITIONAL HINGE-LINE ALONG THE COASTAL PLAIN OF ISRAEL, EVIDENCED BY MAGNETO-TELLURICS

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CONTENTS

page

Abstract 1
Introduction
Purpose of study 1
The longitudinal fault zone of the Coastal Plain – previous works 2
Stratigraphy of the Ḥelez-Ashqelon area 5
Magneto-Telluric Measurements
The magneto-telluric method 6
Data analysis 7
Results 7
Discussion and Synthesis
Significance of the magneto-telluric units 10
Location of a hinge-line and configuration of the basement 11
Possible age of the deeper parts of the sedimentary column of the Coastal Plain 12
Thickness and facies changes along the hinge-line 13
Volcanism and erosion 14
Normal rather than strike-slip fault 16
Synthesis 16
Acknowledgements 17
References 17

Figure 1. Index map of Israel Coastal Plain 2
Figure 2. Location map of magneto-telluric stations and some nearby boreholes 2
Figure 3. Middle Miocene faults in the Coastal Plain (after Gvirtzman, 1970) 3
Figure 4. Profile of the observed total magnetic field in Ḥelez-Ashqelon area (data from Domzalski, 1967) 3
Figure 5. Geological and magneto-telluric cross-sections 4
Figure 6. Magneto-telluric field set-up 6
Figure 7. Correlation between depth, based on magneto-telluric and depth based on geological data 9
Figure 8. The longitudinal Coastal Plain fault-zone 12

Table 1. Results of magneto-telluric measurements at six station in the Ḥelez-Ashqelon area 8
Table 2. Magneto-telluric units 8
Table 3. Correlation between depth and thickness of lithostratigraphic units, and magneto-telluric data 9
Table 4. Summary of geological events along the Coastal Plain hinge-line 15

A STRUCTURAL AND DEPOSITIONAL HINGE-LINE ALONG THE COASTAL PLAIN OF ISRAEL, EVIDENCED BY MAGNETO-TELLURICS

ABSTRACT

An ancient hinge-line, running in a N-S direction occurs in the subsurface of the Israel Coastal Plain. A longitudinal fault zone and stratigraphic changes occur along this hinge-line. A magneto-telluric profile across the fault zone in the Helez-Ashqelon area suggests that the Precambrian basement was displaced from a depth between 6.3 and 8.5 km in the eastern upthrown block, to a depth exceeding 10.6-11.6 km in the western downthrown block. As in the Negev and Sinai, where Cambrian (?), Ordovician, Silurian and Devonian rocks are absent, in the eastern block of the Coastal Plain hinge-line, Carboniferous sediments appear to overlie unconformably the basement. Rocks represented in the sedimentary column of the eastern block are inferred to be Carboniferous-Permian (the Negev Group equivalent) (1.5-2 km), Permo-Triassic (the Ramon Group equivalent) (1.5-2 km), Jurrasic (2-3 km), Cretaceous (approximately 1 km), and Tertiary to Quaternary (0.6 km). The western sedimentary column is inferred to consist of rocks of Infra-Cretaceous (> 8 km), Cretaceous (approximately 1.8 km), and Tertiary to Quaternary age (1.8 km).

Differences in lithology, environment of deposition, and thickness of the Jurassic, Cretaceous and Neogene formations exist between the two sides of the fault zone. The fault is an ancient hinge-line which was active at least in Late Jurassic – Cretaceous and in Neogene times. During the Senonian to Oligocene time interval, when folding movements occurred, the fault was not active. Tectonic activity of the fault is associated with volcanism and in the upthrown eastern block also with erosion. No lateral shift along the fault is apparent.

INTRODUCTION

Purpose of Study

Subsurface data on the Coastal Plain of Israel, based on oil wells and geophysical surveys, provide new information on the stratigraphy and structural geology of Israel. From this information, it was established that both in the Western Coastal Plain and in the Mediterranean shelf, a sequence of shale and marlstone occurs, which is 3-4 km in thickness and which ranges in age from Cretaceous to Quaternary. In the Eastern Coastal Plain, and in the inland area of Israel, however, there is a variable composition of the time equivalent sedimentary column, which includes carbonates, chalks, marlstones, and sandstones. The boundary between the eastern and the western sedimentary provinces seems to coincide with a longitudinal fault zone in the Coastal Plain.

The Coastal Plain of Israel, between Mount Carmel and the Gaza area, is a narrow belt of plains and hills, parallel to the Mediterranean coast. The belt is 5-30 km wide and 0-150 m high (Fig. 1).

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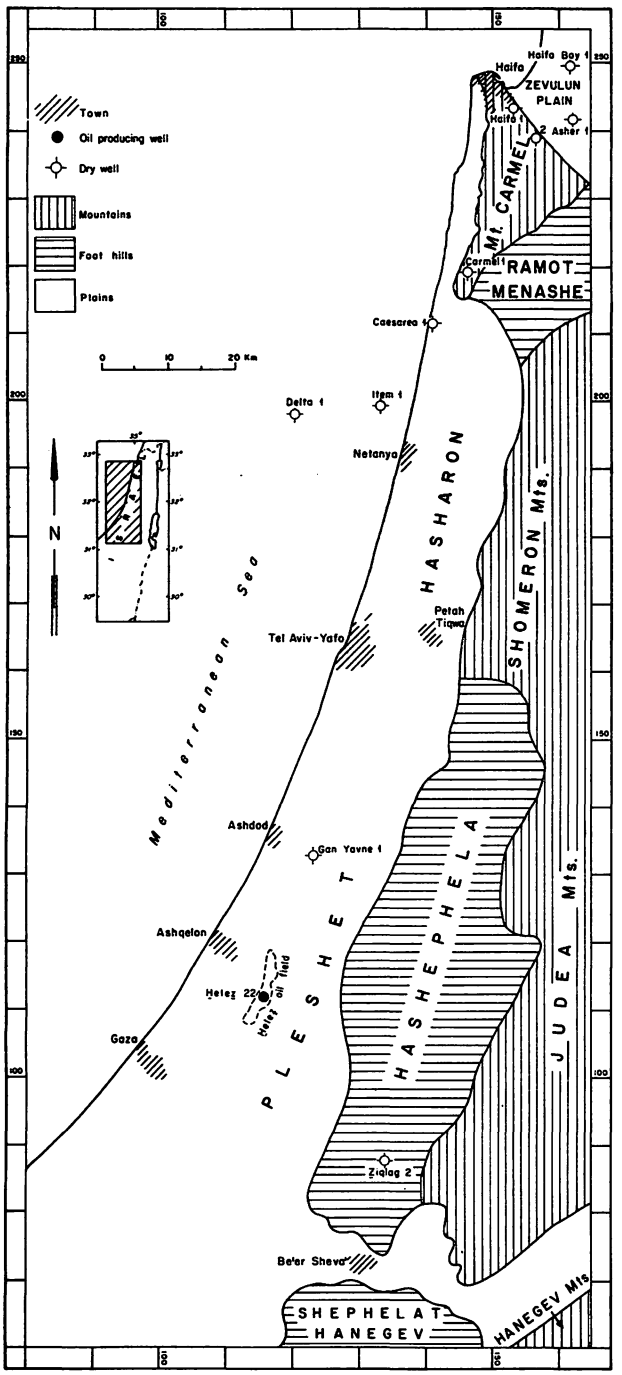


Figure 1. Index map of Israel Coastal Plain

Stratigraphical and structural data of the Coastal Plain indicated the presence of a possible structural and depositional hinge-line, existing since the Jurassic.

In order to prove this hypothesis, a preliminary survey of a magneto-telluric profile, perpendicular to the longitudinal fault lines, was carried out (Fig. 2). The Helez-Ashqelon area was chosen because of abundant available well data. The magneto-telluric method was selected due to deep penetration of measurements it affords. These measurements might indicate the basement configuration of the Coastal Plain.

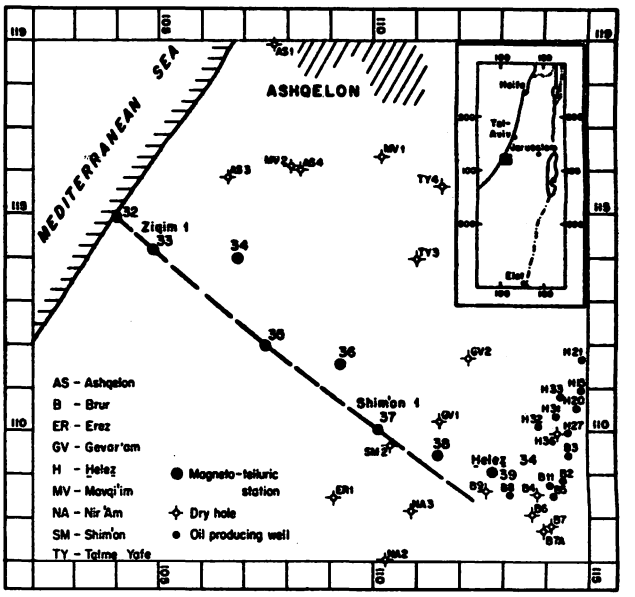


Figure 2. Location map of magneto-telluric stations and some nearby boreholes

The Longitudinal Fault Zone of the Coastal Plain – Previous Works

Picard (1959) named the western downthrown block in the Ashqelon area the "Ashqelon Trough". Gvirtzman (1969a) mapped the fault zone in a series of 1:100,000 subcrop geological maps. According to these maps, the fault zone is composed of several steps downthrown to the west, and is associated with minor transversal faults, running in a WNW direction (Fig. 3). The vertical displacement of the longitudinal faults is several hundreds of meters, and the cumulative displacement of the step faults, according to the base Saqiye datum, is approximately 2 km. The vertical displacement of the transversal faults is several tens of meters.

Mapping of the faults is more definite in the Helez-Ashqelon area because of good well control. In other parts of the Coastal Plain, location of the faults is tentative, and based on interpolation between wells spaced several kilometers apart. Using gravity and seismic data, Ginzburg (1960) mapped a major N-S fault west of the

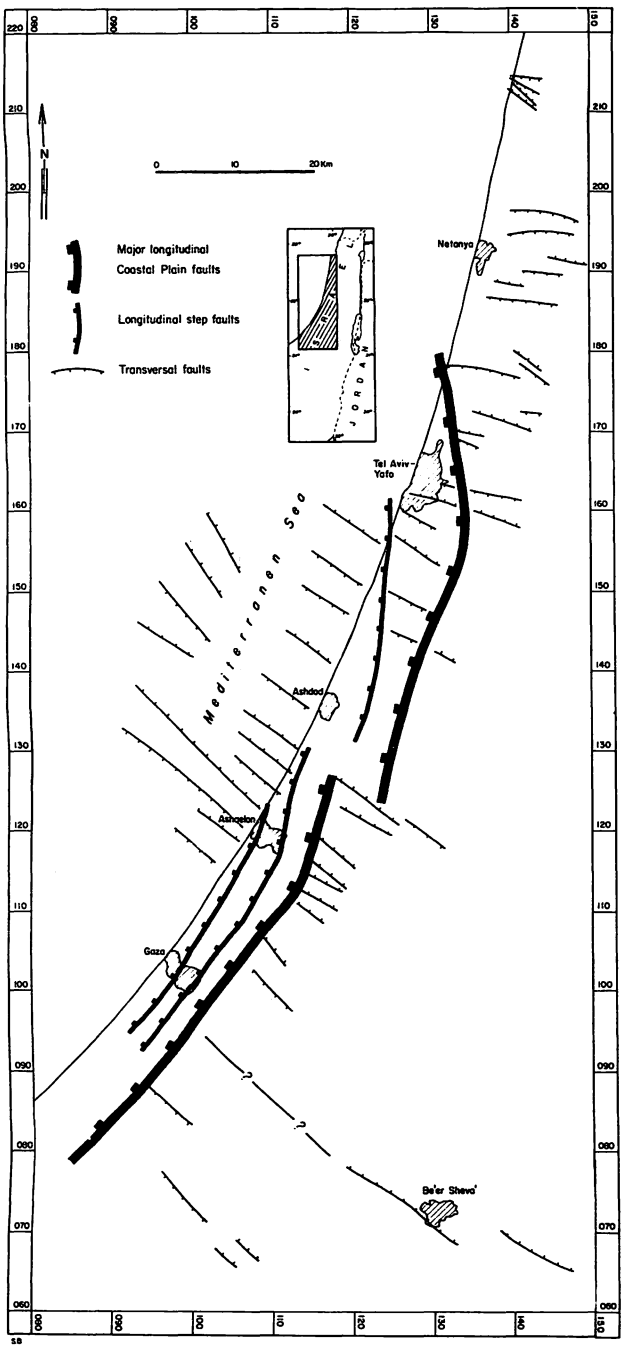


Figure 3. Middle Miocene faults in the Coastal Plain (after Gvirtzman, 1970)

Petaḥ-Tiqwa structure. This fault makes up part of the fault zone discussed in the present study. Also using seismic data, Ginzburg and Amitai (1967), found two parallel longitudinal faults forming steps in the Ashqelon area. Klang (1967) mapped several seismic reflecting horizons of the Gan Yavne area, and showed two parallel steep gradients in a N-S direction. Gvirtzman (1969a) interpreted these two lines as part of the longitudinal fault zone of the Coastal Plain. Greenfield and Bein (in preparation) reinterpreted the seismic data of the Gan Yavne area, and found additional evidence for the existence of the above mentioned faults. Fig. 3 shows the Miocene faults.

The gravity map of the Coastal Plain (Ginzburg, 1960, and new additional data by the Institute for Petroleum Research and Geophysics) shows a steep gradient of decreasing gravity towards the west in the Coastal Plain. This gradient does not specifically indicate the location of the longitudinal fault or fault system, but principally the shape of the fill of the Saqiye Group. The main density contrast occurs between the Saqiye marlstones, and the underlying limestone (Ginzburg, 1960).

The aero-magnetic map of the Coastal Plain (Domzalski, 1967) also indicates a steep gradient of the total magnetic field towards the west (Fig. 4). In his interpretation, Domzalski points out the presence of some longitudinal faults parallel to the present sea shore. However, no significant regional fault zone is indicated.

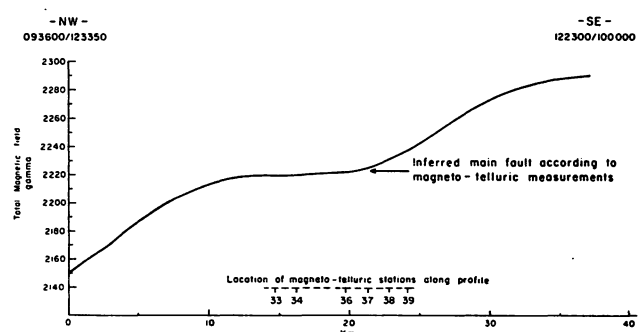


Figure 4. Profile of the observed total magnetic field in Ḥelez-Ashqelon area (data from Domzalski, 1967)

Fig. 5 shows a geological cross-section along with the magneto-telluric profile. This figure shows the approximate thicknesses of the formations, but in detail thickness variations are more complex, as a result of a complex pattern of channelling in the Gevar'am Formation (Cohen, 1969) and in the Saqiye Group (Gvirtzman, 1970).

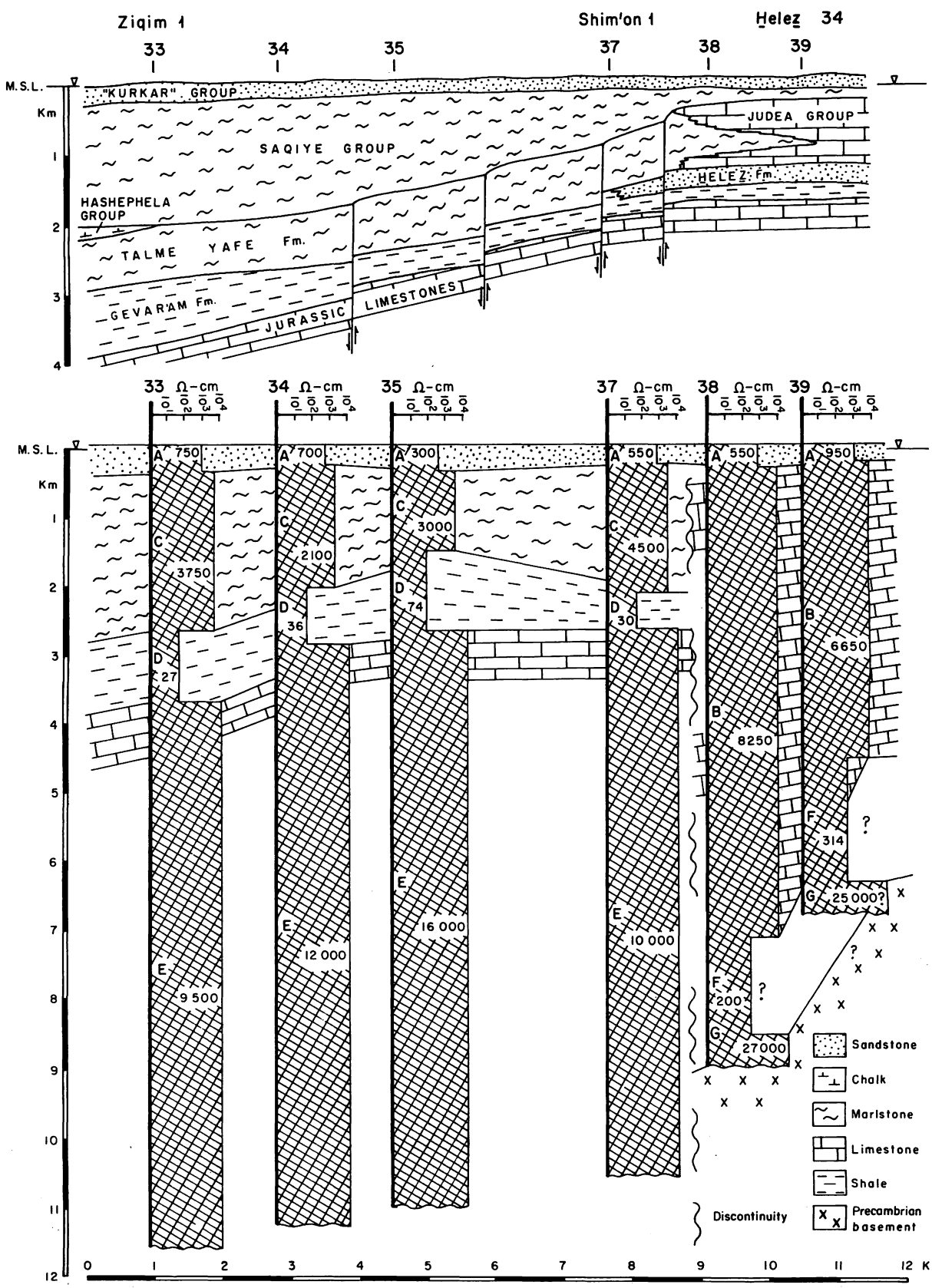


Figure 5. Geological and magneto-telluric cross-sections

Geological data after Cohen (1969), Gvirtzman (1969a), Shenhav and Shoresh (1972). Geophysical data after Ginzburg and Amitai (1967), and Dunkelmann (1968). Magneto-telluric resistivity values are plotted on a logarithmic scale. Units are designated by letters A, B, C, D, E, F, G (Table 2) and by resistivities in ohm-cm. Horizontal = vertical scale.

Stratigraphy of the Ḥelez-Ashqelon area

To understand the geological background as well as the present interpretation of magneto-telluric measurements, the stratigraphic column is reviewed according to well data. This review refers mostly to the Ḥelez-Ashqelon area, as information from that area is particularly abundant. Nevertheless the following description of the main lithostratigraphic units, from top to bottom, is effective for the entire Coastal Plain.

1. **"Kurkar" Group** – (Pleistocene). Composed of calcareous sandstones exposed over the entire Coastal Plain. Thickness ranges from 120m in the east to 160m in the west (Gvirtzman, 1969a, 1969b).
2. **Saqiye Group** – (Late Eocene to Early Pleistocene). Composed of marine marlstone, and distributed throughout the Coastal Plain. To the east, near magneto-telluric stations 38 and 39 (Fig. 2), the thickness is 100-300m. There is a distinct thicknening towards the west, and maximum thickness of 1800m occurs at the present shore line (Gvirtzman, 1969a, 1970).
3. **Hashephela Group** – (Senonian to Middle Eocene). Composed of chalks and marlstones. Sediments of the group were never deposited in the center of the Ḥelez-Ashqelon area. A thin veneer of Middle Eocene chalks covered the center of the area, but was eroded west of it, prior to the deposition of the Saqiye Group. Only minor remains of Hashephela Group still occur in the margins of the Ḥelez-Ashqelon area (Fig. 5) (Gvirtzman, 1969a, 1970).
4. **Judea Group** – (Albian-Turonian). The Judea Group was deposited only in the eastern part of the Coastal Plain, in the area of magneto-telluric stations 38 and 39 (Fig. 5). The sediments are composed of dolomites and limestones which were deposited in a shallow shelf, and are associated with rudist reefs. Thickness is approximately 900m. Towards the west, a facies change into marlstones and chalks of the Talme Yafe Formation occurs. An interfingering zone between the Judea and Talme Yafe sediments was noticed (Fig. 5) (Gvirtzman and Buchbinder, 1966; Arkin and Hamaoui, 1967; Gvirtzman, 1970; Bein, 1970).
5. **Talme Yafe Formation** – (Albian-Cenomanian, ? Turonian). This formation was only deposited in the Western Coastal Plain, and is composed of marlstone and chalks which were deposited under deeper marine conditions contemporaneously with the Judea Group. Thickness is approximately 900m (Gvirtzman and Buchbinder, 1966; Cohen, 1969; Gvirtzman, 1970; Bein 1970, 1971).
6. **Ḥelez Formation** – (Valanginian to Barremian). The formation is composed of sandstones, shales, dolomites and limestones, and is only distributed in the Eastern Coastal Plain. Sands of the formation from the reservoir rocks of the Ḥelez-Kokhav oil fields. The Ḥelez Formation was deposited in near-shore environments having both continental and marine elements. The formation changes laterally and downwards into the Gevar'am Formation which was deposited under deeper marine conditions; its thickness is approximately 250m (Cohen, 1969; Shenhav, 1970, 1971).
7. **Gevar'am Formation** – (Barriasian to Valanginian). The formation is composed of dark shale rich in organic matter and hydrocarbons. The formation is considered the source rock of the Ḥelez-Kokhav oil fields. It was deposited under deeper marine conditions; thickness ranges from 100m in the east to nearly 1,000, in the west (Cohen, 1969; Shenhav, 1970, 1971; Shenhav and Shoresh, 1972).
8. **Jurassic rocks** – These consist mostly of limestones, dolomites and some shale, mostly deposited under shallow marine conditions. Maximum known thickness is 2,500m, penetrated in the Ḥelez 22 well (Picard and Eliezri, 1964; Derin, 1966; Cohen, 1969; Barzel and Friedman, 1970). According to B. Derin (personal note) the bottom of the Ḥelez 22 well is close to the Jurassic-Triassic bound-

ary. No deeper formations were penetrated in the Coastal Plain.

For more details the reader is referred to illustrations which compile data based on columnar sections, facies studies and subsurface mapping, as follows: Jurassic in the Ḥelez̲ area – Barzel and Friedman (1970, Fig. 22, p. 205); Jurassic in the Northern Coastal Plain – Friedman, Barzel and Derin (1971, Fig. 3, p. 15); Cretaceous of the Ḥelez̲ area – Shenhav (1971, Fig. 3, p. 2197), Bein (1971, Fig. 7, p. 45); Cretaceous of the Northern Coastal Plain – Bein (1971, Fig. 4, p. 39); Upper Cretaceous and Tertiary in the Ḥelez̲ area – Gvirtzman (1970, Fig. 2, p. 15); Upper Cretaceous and Tertiary in the Southern Coastal Plain – Gvirtzman (1969a, cross sections 5 and 6).

MAGNETO-TELLURIC MEASUREMENTS

The Magneto-Telluric Method

The magneto-telluric method, based on Cagniard's theory (1953) of micropulsations of plane electromagnetic waves incident on the earth, involves simultaneous surface measurement of the horizontal components of the magnetic and electric fields.

Cagniard showed that quantitative information concerning the electrical properties of the subsurface can be obtained for horizontally layered earth, where each layer is homogeneous and isotropic. He demonstrated that resistivity can be found as a function of depth if the apparent resistivity ρ_a is known as a function of period T.

Apparent resistivity is computed from:

$$\rho_a = 0.2T \left[\frac{E_y}{H_x}\right]^2$$

where

ρ_a apparent resistivity in ohm-meters
T period of the magneto-telluric pulsation in seconds
E_y horizontal component of electric field in direction y, in mV/km
H_x horizontal component of magnetic field in a perpendicular direction x, in gamma (γ).

The maximum depth penetration depends on the longest period measured in the field. In the used system, only periods up to 200-300 seconds could be measured. Thus, the maximal depth penetration obtained in the present study is about 8-12 km.

In the local application of the magneto-telluric method one horizontal component of the electric field (E_y) and one of the magnetic field (H_x) were measured (Fig. 6).

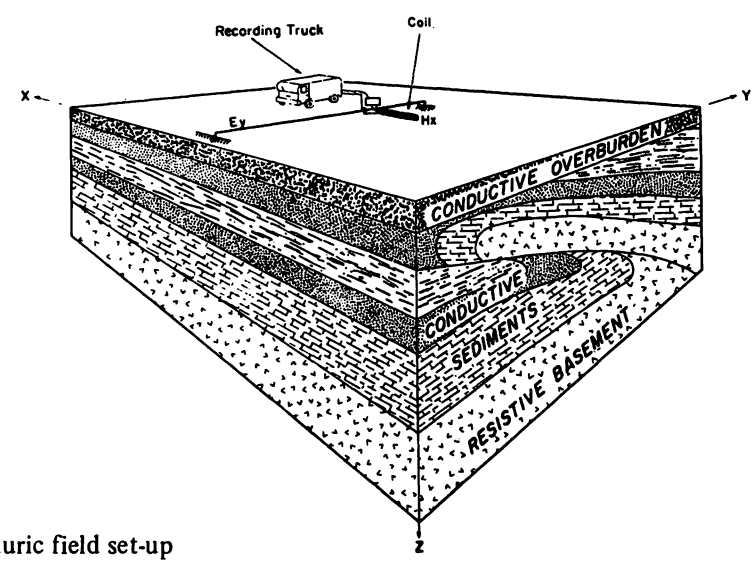


Figure 6. Magneto-telluric field set-up

The magnetic system consists of a large induction coil whose signal is amplified by a three-stage amplifier with a calibrated gain. Included in the amplifier are notch filters for 50 Hz and 150 Hz and a system of band pass filters (combination of high-cut and low-cut filters) for the pre-filtering of signals. The telluric system (measuring the electric field) consists of two porous-pot electrodes planted in the ground, about 600m apart connected by insulated leads to the same amplifier as that used for the magnetic system. This recording system was built by "Centre d'Etudes Géophysiques", France.

Data Analysis

Eight stations were chosen for field measurements, the locations of which are shown in Fig. 2. The stations are located in a line perpendicular to the faults of the Coastal Plain. Data obtained at station 36 could not be used, and this station has not been considered in the discussion below. Data obtained at station 32 show unusually low resistivity values at a depth of about 2.5 km. Close to this station there are some boreholes and seismic shot points. No geological or geophysical features occur that might be connected with the low resistivity values. It is assumed, therefore, that the results are due to instrumental interference. These results do not present the lithology of the strata, and this site too, was not included in the following discussion.

Records obtained directly, or by play-back of the magnetic tapes for the chosen periods in the period band measured were analyzed individually, and all the correlated magnetic and telluric events were considered. From the ratio of amplitudes of these events the values of the apparent resistivities ρ_a were calculated for the chosen periods.

Values of ρ_a versus $\sqrt{T}$ were plotted on log-log coordinates, and using master curves, the best fit model of resistivities and thickness was calculated. Using a multilayer case computer program, this proposed model was verified by reconstructing the corresponding theoretical curve.

Results

Table 1 lists the results of magneto-telluric measurements of six stations along a profile in the Ḥelez-Ashqelon area, perpendicular to the fault zone (Fig. 2). The results show depth soundings of intervals measured in meters and characterized by resistivity values in ohm-cms. Fig. 5 shows a geological cross-section (horizontal = vertical scale) along the measured profile, indicating the main lithostratigraphic units down to the jurassic.

At the bottom of Fig. 5 the results of the magneto-telluric measurements are shown along the same profile. The resistivity values are plotted on a log scale. The magneto-telluric units are designated by letters as magneto-telluric units. Table 2 lists the resistivities and thicknesses of the magneto-telluric units and their possible lithostratigraphic equivalents. From the cross-sections of Fig. 5, there appears to be good correlation between Unit A and the "Kurkar" Group, between Unit C and the Saqiye + Talme Yafe formations and between Unit D and the Gevar'am Formation. Table 3 shows details of these correlations. Although there are differences between the depth and boundary of the lithological and geophysical units, a statistical analysis of correlation between the 4 contacts obtained, shows a correlation coefficient as high as 0.96 (Fig. 7).

Table 1. Results of Magneto-telluric measurements at six stations in the Helez-Ashqelon area

Station	Coordinates	Elevation, MSL	Interval, m.	Thickness, m.	Resistivity, Ω - cm	Unit Symbol
33	104800/114270	22.0	0 - 290	290	750	A
			290 - 2,600	2,310	3,750	C
			2,600 - 3,640	1,040	27	D
			3,640 - > 11,600	> 7,960	9,500	E
34	106800/114000	25.0	0 - 160	160	700	A
			160 - 2,000	1,840	2,100	C
			2,000 - 2,800	800	36	D
			2,800 - > 11,300	> 8,500	12,000	E
35	107500/111950	25.0	0 - 290	290	300	A
			290 - 1,450	1,160	3,000	C
			1,450 - 2,620	1,170	74	D
			2,620 - > 11,100	> 8,480	16,000	E
37	110054/110075	51.0	0 - 190	190	550	A
			190 - 2,100	1,910	4,500	C
			2,100 - 2,600	500	30	D
			2,600 - > 10,600	> 8,000	10,000	E
38	111500/109400	75.0	0 - 320	320	550	A
			320 - 7,100	6,780	8,250	B
			7,100 - 8,500	1,400	200	F
			8,500 -		27,000	G
39	112750/109000	75.0	0 - 280	280	950	A
			280 - 4,500	4,220	6,650	B
			4,500 - 6,300	1,800	314	F
			6,300		? 25,000	G

Table 2. Magneto-telluric units

Magneto-telluric unit	Resistivity range ohm - cm	Thickness range km	Inferred equivalent lithostratigraphic unit	Inferred lithology and remarks
A	300 - 950	0.160 - 0.320	"Kurkar" Group	Calcareous sandstones, exposed, the upper part is a fresh water aquifer
B	6,650 - 8,250	4.220 - 6.780	Judea Group (including a tongue of the Talme-Yafe Formation), Helez Formation (including wedge of the Gevar'am Formation) Jurassic carbonates and older sediments.	Predominantly hard limestones and dolomites
C	2,100 - 4,500	1.160 - 2.310	Saqiye Group and Talme-Yafe Formation	Predominantly marlstones
D	27 - 74	0.500 - 1.170	Gevar'am Formation	Very low resistivity - shale
E	9,500 - 16,000	7.960 - 8.500(+)	Jurassic carbonates and older sediments	Probably hard carbonates
F	200 - 314	1.400 - 1.800	?	? Shale, Probably overpressured
G	25,000 - 27,000	?	Precambrian Basement	Probably igneous and metamorphic rocks

Table 3. Correlation between depth and thickness of lithostratigraphic units, and magneto-telluric data

STATIONS / UNITS		33 (near Ziqim 1 well)		34		35		37 (near Shim'on 1 well)		38		39 (near Helez 34 well)	
Lithostratigraphic and Magneto-telluric		by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.	by geology*	by mag.-tell.
"Kurkar" Group, "A"	bottom, m.	167	290	140	160	130	290	91	190	125	320	125	280
	thickness, m.	167	290	140	160	130	290	91	190	125	320	125	280
Saqiye Group and Talme-Yafe Formation, "C"	bottom, m.	2775	2600	2450	2000	2300	1450	1614	2100	Only negligible thickness of Saqiye and no Talme-Yafe in this area			
	thickness, m.	2608	2310	2310	1840	2170	1160	1523	1910	No Unit "C"			
Gevar'am Formation, "D"	bottom, m.	3770	3640	3200	2800	2750	2620	1924	2600	Only negligible thickness of Gevar'am Formation in this area			
	thickness, m.	965	1040	750	800	450	1170	310	500	No Unit "D"			

* including by seismic data

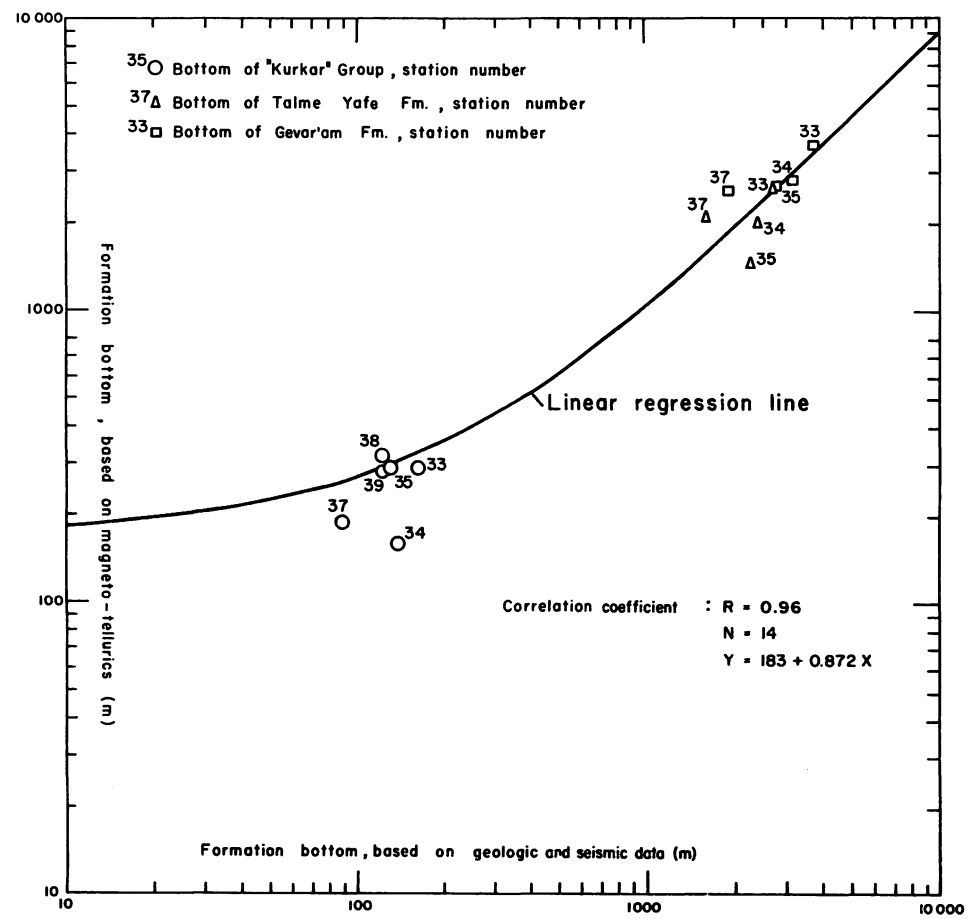


Figure 7. Correlation between depth, based on magneto-telluric and depth based on geological data

DISCUSSION AND SYNTHESIS

Significance of the magneto-telluric units

For the purpose of the present discussion, a magneto-telluric unit is defined by its resistivity and thickness ranges and by its position in the sequence. Table 2 is a summary of the properties of the defined magneto-telluric units. Fig. 5 shows the same units in cross-section. The following are complementary remarks to Table 2 and Fig. 5.

Unit A – This Unit is exposed, its upper part being within the weathered zone or the "conductive overburden" (Fig. 6). In spite of relatively low resistivity values, the upper part of this unit makes up the fresh water aquifer of the Coastal Plain.

Unit B – This unit is only found in the eastern part of the section (stations 38 and 39, Fig. 5). The unit differs from Unit C, which forms the lateral extention of B to the west, in that resistivity values of Unit B are double those of Unit C (Table 2). Unit B has a thickness of 4-7 km, whereas Unit C is only 2 km in thickness. The underlying units also differ; B overlies F, and C overlies D. D and F are different units with different properties. The wells of the Ḥelez-Kokhav field penetrate the interval of the upper part of Unit B. This depth interval includes the approximate 900m of the Judea Group carbonates, the approximate 250m of the Ḥelez Formation sandstones, the approximate 100m of the wedged-out Gevar'am Formation shale and the approximate 2.5 km of Jurassic carbonates which were penetrated in the Ḥelez 22 well. The total section mentioned above (approximately 3.75 km), is mostly composed of carbonates. The approximately 250m of sandstone and shale do not change the geophysical properties of the entire Unit B sequence. The lower part of Unit B is older than Jurassic, but is probably of similar lithology.

Unit C – This unit is the lateral westward extention of the upper part of Unit B. It corresponds to the marlstones of the Saqiye Group, and the Talme Yafe Formation.

Unit D – This unit is equivalent to the Gevar'am Formation. Extremely low resistivity values were recorded from this unit. According to Fertl and Timko (1970), very low values may indicate overpressured conditions in which the formation pressure exceeds the expected hydrostatic pressure. However, pressures measured in wells that penetrated the Gevar'am Formation are not abnormally high.

Unit E – This unit is the thickest in the studied area. The bottom of the unit was not reached at 8 km. The upper part of the unit corresponds to the Jurassic carbonates which were penetrated in wells in the Ashqelon area. The entire unit is apparently of carbonates.

Unit F – This unit is only known in the eastern part of the profile. It is possible, however, that this unit may underlie Unit E in the west. Resistivity values of F are low.

Unit G – Due to its very high resistivity, this unit is presumed to be the Precambrian basement. In other magneto-telluric measurements in Israel, resistivities of Precabmrian basement rocks were found in the order of magnitude of 18-46 x 10^3 ohm-cm. Measurements in the Timna' area in the Negev (Klang, 1970) showed the resistivity of the crystalline Precabmrian basement to be 18 x 10^3 ohm-cm.

In the Ẓenifim-Paran-Meishar area in the Negev (Weissbrod and Klang, in preparation) values of 19-46 x 10^3 ohm-cm were recorded at a depth that corresponds to the so-called "Infra-Cambrian Ẓenifim Arkose" (Weissbrod, Personal note, see also Weissbrod, 1969a). The Ẓenifim Arkose, a thick sequence in the Northern Negev, composed of degradation granitic particles, is part of the so-called "Precambrian Basement" of the Arabo-Nubian massif, which is mainly composed of igneous and metamorphic rocks. In Israel, therefore, magneto-telluric data point to the Ẓenifim Formation as part of the Precambrian basement complex. When thus classifield, the prefix "Infra-Cambrian" which was previously given to the formation can be disregarded. In the other magneto-telluric measurements that were carried out in Israel, such high

resistivities were not found within the overlying sedimentary column (except in the salt-beds of the Dead Sea area). Consequently, the G unit of the Ḥelez area is correlated with the Precambrian basement. This type of unit was not penetrated in stations at the western part of the profile.

Location of a hinge-line and configuration of the basement

For the purpose of the present discussion, an ancient hinge-line is defined, as elongated zone along which fault displacement, sharp changes in thickness of sediments and/or sharp changes in environments of deposition have occurred. A hinge-line of this nature is probably the expression of an active fracture in the earth's crust, known to cause rejuvenation of ancient faults. The latter process brings about paleogeographic features such as shelf edges, fault scarps, etc. These paleogeographic features form boundaries between sedimentary provinces and different environments of deposition. Thus, the term "hinge-line", in this discussion, indicates a tectonic and/or a depositional hinge. The hinge-line seems to be several kilometers in width and includes the entire zone of step faults and facies interfingering.

Magneto-telluric measurements in the Ḥelez-Ashqelon area (Fig. 5) indicate a major lateral change in the nature of the magneto-telluric units between stations 37 and 38. This lateral change is even more apparent if we compare stations 37 and 39. The geological cross-section also shows a lateral change in the upper sedimentary column, mostly between stations 37 and 38. The Ḥelez Formation passes laterally to the Gevar'am Formation, the Judea Group passes to the Talme Yafe Formation and from a negligible thickness of the Saqiye Group, there is a change into a thick section of the Saqiye Group.

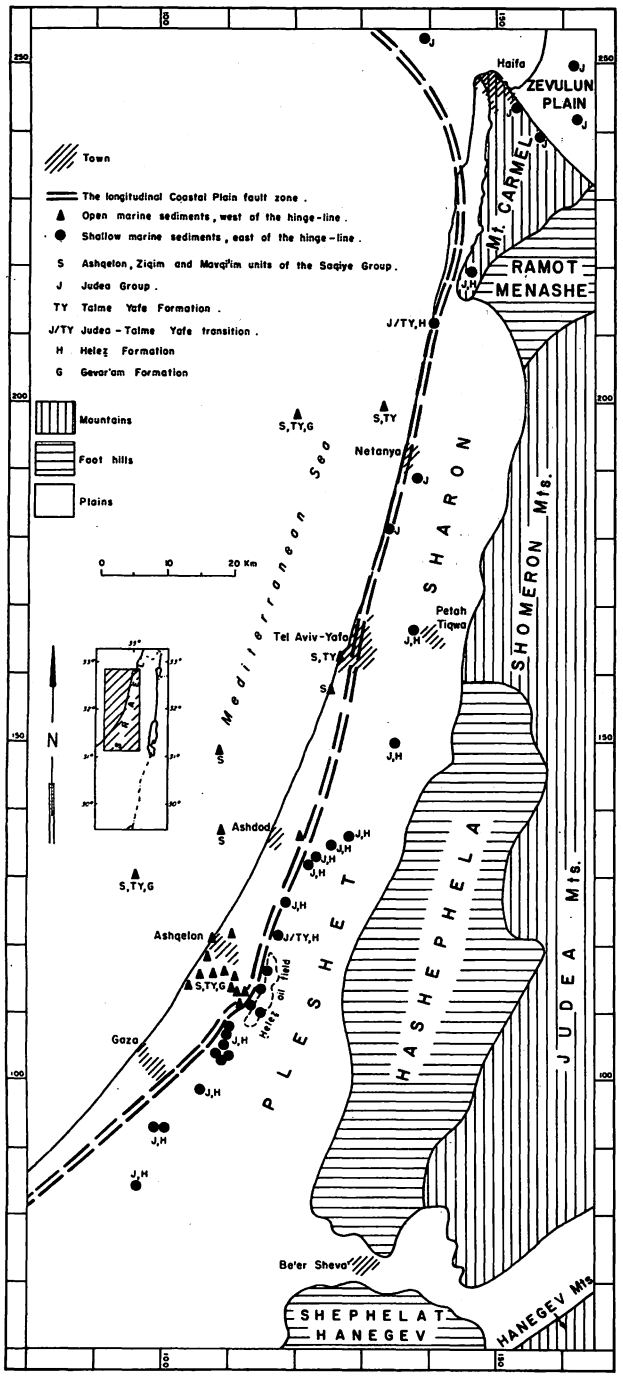
In stations 38 and 39, the depth of the Precambrian basement is found to be of 6.3 and 8.5 kms (Tables 1 and 2). There is a gradient to the west between these two stations, which might express a fault or a regional dip. The Precambrian basement was not penetrated in the next four stations but it is deeper than 10.6 km in station 37 and deeper than 11.6 km in station 33, the westernmost station. The fact that the Precambrian basement was not penetrated in station 37, might support the idea that the main fault zone is located between station 37 and 38, or more generally, a basement step fault zone might exist between stations 37 and 38. This assumption is equally supported by an aero-magnetic profile. The aero-magnetic map here under discussion (Domzalski, 1967) indicates two steps in the profile of the observed total magnetic field along the direction of the magneto-telluric survey (Fig. 4). The main fault zone, between stations 37 and 38 is a boundary between the steps in the magnetic profile.

In conclusion, the above-mentioned changes are expressed by the following features: a lateral change in the magneto-telluric properties of the sedimentary column; a lateral facies change in the upper sedimentary column; a fault zone, and a cumulative displacement of the precambrian basement of more than 3-5 km; and a steep gradient in the total magnetic field along the profile. Most of these features are focussed between stations 37 and 39.

Considering the change of the width of the hinge-line belt during geological times, one might deduce that the later the time the wider is the hinge-line belt. The inferred displacement zone of the basement is situated within a distance of 3 km, between stations 37 and 39. The width of the interfingering zone between the Ḥelez and the Gervar'am Formations and between the Talme Yafe and the Judea units, in the Ḥelez area, is approximately 4-5 km (the cross-section of Fig. 5 shows a distance of about 3 km, however, the comm on width in the entire area, according to Cohen, 1969, is about 4-5 km). According to Bein (1971, Fig. 4, p. 39) the width of the Judea-Talme Yafe interfingering zone in the Northern Coastal Plain is about 5-6 km. The width of the Miocene step fault belt, according to Gvirtzman (1970), is 8-10 km (see Fig. 3).

It appears that the fault zone is actually an

old fracture in the earth's crust. The narrow fracture in the basement becomes wider up in the sedimentary column. This fracturing of the crust was active several times during the geological history of the Coastal Plain. This hypothesis is discussed in the following paragraphs. The estimated location of the Coastal Plain fault zone or hinge-line is shown in Fig. 8.



← Figure 8. The longitudinal Coastal Plain fault-zone

Possible age of the deeper parts of the sedimentary column of the Coastal Plain

In wells of the Northern Negev, as well as in outcrops of southwest Sinai Carboniferous rocks unconformably overly "Infra-Cambrian" arkosic rocks (Zenifim Formation) and/or thin marine Cambrian sediments. In places Carboniferous rocks also directly overly the crystalline basement. In the Negev as well as in Sinai there is a huge sedimentary gap, in which Ordovician, Silurian and Devonian sediments are lacking. These Lower Paleozoic sediments are only found in northwest Saudi Arabia and in south Jordan. In the Negev and in Sinai the Carboniferous sediments overly a widespread truncated relief (Weissbrod, 1969a, 1969b, 1970).

For the most part, the thickness of the sedimentary cover which overlies the truncated old relief in the Northern Negev may be divides as follows: 800m of the Carboniferous – Permian Negev Group (Weissbrod, 1969a), 800m of the Permo-Triassic Ramon Group (Druckman, 1966) and 1250m of the Jurassic Arad Group (1000m from bottom of the section to the top of Zohar Formation according to Goldberg, 1970, and approximately 250m of the Jurassic Kidod, Be'er Sheva' and Haluza formations). The above-mentioned groups thicken towards the northwest, in the general trend from the Arabo-Nubian massif into the old Tethys ocean. The rate of thickening towards the Helez area in the Coastal Plain may be estimated according to the thickening of the Jurassic sequence. (The Helez area, in this discussion, is only the eastern block of the inferred hinge-line.) In the Ziqlag 2 well, between the Northern Negev and the Coastal Plain, 1620m of Jurassic werepenetrated (Fisher and Druckman, 1968); in the Helez 22 well, 2500m of Jurassic were penetrated. According to B. Derin (personal communication), the bottom of the penetrated sequence in the Helez 22 well is very close to the Jurassic-Triassic boundary. Thus it may be concluded that the Jurassic of the Coastal Plain is approximately twice as thick as the Jurassic of the Negev. A twofold thickening is also assumed for the Negev and

Ramon groups and the expected section in the Ḥelez area is 1600m of Carboniferous – Permian, 1600m of Permo-Triassic in addition to the 2500m of Jurassic. A total of 5.7 km of Carboniferous-Jurassic sediments is estimated.

The magneto-telluric measurements in the eastern part of the profile (station 38 and 39) indicate similar thicknesses. In station 39, which is located near the Ḥelez 34 well and in which the top of the Jurassic is at 1760m depth, the total thickness of Infra-Cretaceous sediments is 4.5 km (from 1760m to 6300m, the top of the inferred Precambrian basement). In station 38 where the top of the Jurassic, by extrapolation and seismic data, is at 1800m depth, the total thickness of Infra-Cretaceous sediments is 6.7 km (from 1800m to 8500m, the top of the inferred Precambrian basement). The thickness range of the Infra-Cretaceous sediments in stations 38 and 39 is 4.5-6.7 km, which is close to the above-mentioned estimated 5.7 km of Carboniferous-Jurassic sediments. From the similarity of these estimates a working hypothesis may be deduced that the total Infra-Cretaceous sediments in the Ḥelez area is also of late Paleozoic to Mesozoic age. Lower Paleozoic sediments, which are absent in the Negev and Sinai (either due to non deposition or truncation) are probably also absent in the eastern block of the Coastal Plain belt.

Thickness and facies changes along the hinge-line

Infra-Cretaceous – For the purpose of the present discussion, Infra-Cretaceous is defined as Jurassic or older. However, it is probable, as assumed previously, that the Precambrian basement, in the upthrown block of the Coastal Plain, is overlain by Carboniferous rocks. The Infra-Cretaceous sediments in the east, therefore, are probably Carboniferous to Jurassic.

As mentioned previously, the thickness range of the Infra-Cretaceous sediments in the eastern block is 4.5-6.7 km. In the western downfaulted block, in station 37, the thickness of the infra-Cretaceous sediments (Unit E), exceeds 8 km and the thickness in station 34 exceeds 8.5 km. The thicker section of the Infra-Cretaceous in the western downthrown block is either due to the presence of sediments older than Carboniferous, or to a thicker accumulation of sediments dating from the Carboniferous to the Jurassic time interval.

There are some indications for facies changes in Jurassic sediments. There is no sharp change in lithology in Jurassic carbonates between the two sides of the hinge-line in the Ḥelez-Ashqelon area. Nevertheless, Barzel and Friedman (1970) found different environments of deposition of late Jurassic sediments between the carbonates of the Ḥelez belt, which were deposited in very shallow marine conditions, in a high-energy zone above the sea wave base, and those west of the Ḥelez belt, which, they inferred, were deposited under low-energy open sea conditions, beneath wave base. In the Northern Coastal Plain, along the northern extention of the fault zone, there is a distinct difference in Jurassic rocks on either side of the hinge-line. Friedman, Barzel and Derin (1971) found different environments of deposition in the Upper Jurassic rocks. In the eastern upthrown block of the fault zone, as suggested in the present study, which is represented by the Caesarea 1, Carmel 1, Haifa 1, Asher 1 and 2, and Haifa Bay 1 boreholes, they found Jurassic sediment of basin margin (seawards, off the basin slope) environment.

Facies changes in the late Jurassic sediments might be connected with a displacement in the fault zone and with an accumulation of thicker sediments in the downthrown block. It is also possible that the few kilometers of thicker section in the western block, were accumulated during the Jurassic rather than over the entire pre-Cretaceous to post-Precambrian time interval.

Lower Cretaceous – A facies change also occurs in Lower Cretaceous sediments on either side of the hinge-line. The Ḥelez Formation in the eastern block includes sediments of continental dune sands, tidal channels and lagoons, offshore marine sands as well as reef dolomites, carbonaceous shale and volcanic ash (Shenhav,

1970, 1971). The formation passes laterally into the Gevar'am Formation, which is mostly distributed along the western side of the hinge-line. The Gevar'am Formation was deposited in open sea conditions (Cohen, 1969). Shenhav and Shoresh (1972) pointed out some evidence for a rapid rate of deposition of the Gevar'am Shale, which explains the accumulation of more than 950m of the shale in the downfaulted block, while the Ḥelez Formation, which was deposited near the shore has a thickness of only 250 m.

In the western downthrown block an excess of sediments of approximately 0.7 km was deposited during early Cretaceous times.

Albian-Cenomanian – In rocks of Albian to Cenomanian age, there is also a facies change on either side of the hinge-line. The Judea Group, in the eastern block is composed of shallow marine carbonates, rudist reefs, and platform limestones and dolomites (Bein, 1971). To the west, there is a facies change through a zone of interfingering, into open sea marlstones (Gvirtzman and Buchbinder, 1966; Cohen, 1969; Gvirtzman, 1970; Bein, 1970, 1971).

There is no distinctive difference in thickness between the Judea Group and the Talme Yafe Formation in the Ḥelez-Ashqelon area. Nevertheless, 2.6 km of Albian to Cenomanian sediments were penetrated in the Item 1 off-shore well, which faces the Northern Coastal Plain, and which represents the downthrown block. However, thickness of the Judea Group, in the upthrown block of the Northern Coastal Plain is only 700-900m.

Neogene – In sediments of the Saqiye Group (late Eocene to early Pleistocene) there is also a distinct facies change between the two sides of the hinge-line. In the western downthrown block, marine conditions persisted almost throughout the entire time of deposition of the Saqiye Group (except during a relatively short break in the Miocene) and approximately 1800m of open marine globigerinal marlstones were deposited. The Ashqelon Member, and the Ziqim and Mavqi'im Formations were only deposited in the western trough.

The eastern upthrown block was exposed to erosion during the early to middle Miocene times. Along the hinge-line there was a shore belt in which the fault scarp formed a sea-shore cliff. The eastern block was an exposed inland plateau, where erosion channels were cut into the fault scarp (Gvirtzman, 1969a, 1970). The cumulative displacement of the fault zone during the Neogene was estimated as approximately 2 km (Gvirtzman, 1970).

Rates of displacements – Following the above hypothesis it is possible to estimate rates of displacements between the two sides of the hinge-line in the Ḥelez-Cretaceous area. It is assumed that the "Infra-Cretaceous displacement" took place mostly in the late Jurassic times and is associated with the facies changes of these times. The Lower Cretaceous and Albian-Cenomanian displacements seem to be a continuation of the same movement. The movement probably commenced during the late Jurassic and probably terminated during the late Cenomanian. The excess of thickness of Infra-Cretaceous sediments in the western block is more than 1.8-4.0 km. During the Lower Cretaceous, 0.7 km of sediments were accumulated. No distinctive accumulation differences are known during the Albian-Cenomanian. As a possible result of Jurassic-Cretaceous movements, the total excess of sediments in the western block area, exceeds 2.5-4.7 km. The Jurassic-Cretaceous displacement continued approximately over a period of 70 million years and therefore, the rate of displacement is estimated as over 36-67m per one million years.

The Neogene displacement is estimated as 100m per one million years (Gvirtzman, 1970). A graphic summary of these estimations is presented in Table 4.

Volcanism and erosion

Tectonic movements in the fault zone along the hinge-line area, both during the late Jurassic-Cretaceous, and during the Neogene, appear to be also connected with volcanic activities and erosional phases in the Coastal Plain.

In Israel the Jurassic-Cretaceous movement

Table 4. Summary of geological events along the Coastal Plain hinge-line

Time in million years	Series	Cummulative vertical displacement in kilometers versus time in Million years (Km 1 2 3 4 5)	Differential vertical displacement, Kms	Cummulative vertical displacement, Kms	Volcanic activities	Major unconformities	Sediments of the western downthrown block: Formation or Group	Common thickness, m	Lithology	Environments of deposition	Sediments of the eastern upthrown block: Formation or Group	Common thickness, m	Lithology	Environments of deposition
50	TERTIARY		2.0	>(4.5-6.7)	Galilee Volcanics, National Park Volcanics		Yafo Fm.	550	marlst.	open marine	Yafo Fm.	100	marlst.	open to shallow marine
							Mavqi'im Fm.	150	anhydr.	lagoonal ?				
							Ziqim Fm.	200	marlst.	open deeper marine				
							Ashqelon Mbr.	250	marlst.					
100	CRETACEOUS		0.7	>(2.5-4.7)	Volcanics in Carmel		Talme - Yafe Fm.	800	marlst. & chalks	open marine	Judea Group	800	limest. & dolomite	shallow shelf, rudist reefs
						?	Gevar'am Fm.	700	shale	open deeper marine	Helez Fm.	200	sands, shale, carbona.	Near shore, tidal, beach, eolian
150	JURASSIC	?	>(1.8-4.0)	>(1.8-4.0)	Volcanics in Israel		unnamed formations	?	limest.	in Ashqelon: low energy, shallow in Carmel: outer slope	unnamed formations	?	limest. and dolomite	shallow marine
	TRIASSIC													

was accompanied by volcanics. Volcanic rocks from the Jurassic-Cretaceous boundary were found in boreholes of the Northern Coastal Plain and Northern Israel (Nachmias and Gvirtzman, 1967). The proximity of the Helez Formation sediments to volcanic rocks is discussed by Shenhav (1970, 1971), while Cretaceous volcanism in the Carmel and Umm el Fahm areas is described by Sass (1957, 1968). Cainozoic volcanic activities in Israel were described by Bentor (1957). Neogene volcanic activities in the Coastal Plain (National Park Volcanics) associated with the fault zone are described by Grader and Gvirtzman (1961) and Gvirtzman (1970).

Erosional phases, which are expressed by major unconformities, have been considered by several authors. The Jurassic-Cretaceous unconformity is discussed by Eliezri (1964) and Cohen (1969). Greenfield and Bein (in preparation) suggested that the erosion channel in the Helez-Ashqelon area, which cut the Jurassic rocks, and was filled by the Lower Cretaceous Gevar'am Formation, was formed due to erosion in the rising upthrown block. The major Neogene uplifting, erosion and unconformity is discussed by Gvirtzman (1970) and illustrated in subcrop geological maps (Gvirtzman, 1969a).

Normal rather than strike-slip fault

Differences in thickness and environment of deposition between two sides of a fault can either be explained, as suggested above, as an ancient hinge-line, or else as a strike slip fault. Since the two sides of the fault do match on opposite sides, as discussed in detail below, it seems that it is not a strike slip fault.

The differences in the Jurassic sediments on both sides of the Dead Sea rift have been previously explained to be a result of an ancient hinge-line (Bentor and Vroman, 1969) but were later considered to be the result of a strike-slip fault, in which both sides of the rift moved 105 km apart (for details see Freund *et al.* 1970).

There are indications that the Coastal Plain fault zone is an ancient hinge-line, at least since the Jurassic, and that it is not a left-lateral offset fault. The lateral facies change between the Gevar'am and Ḥelez Formations is associated with an erosional channel cut into the Jurassic rocks and filled by Lower Cretaceous marine sediments of the Gevar'am Formation. The channel of the Ḥelez area which was mapped in detail (Cohen, 1969; Shenhav and Shoresh, 1972), extends across the fault line. The isochore map of the channel fill by the Gevar'am sediments, shows a gradual change from more than 950m in the western downfaulted side, to a few tens of meters in the eastern upthrown block. The axis of the channel is regular and does not indicate any lateral shift.

The facies change between the Talme Yafe Formation and the Judea Group is through a zone, 4-5 km in width, of interfingering between tongues of the Judea and Talme Yafe units (see Fig. 5). An interfingering zone is a clear indication of true facies changes and cannot be explained to be the result of a later lateral shift. Furthermore, the tongues of Talme Yafe can be followed in detail from the downfaulted to the upthrown blocks (see Shenhav, 1971, Fig. 3 and Bein, 1971, Fig. 7).

During the Miocene, erosion cut deep channels in the upthrown block. The extention of the canyons into the downthrown block can also be followed. Gvirtzman (1969a, 1970) mapped the Miocene drainage system in detail. In this map, which is mostly based on seismic reflections, a regular drainage pattern is shown, without any indication of lateral shifting of the system.

The above arguments show that no lateral displacement has occured since the early Cretaceous and the cause for facies changes, and differences in thicknesses is the vertical displacement of segments of the ancient hinge-line. The hinge-line is, therefore, an old normal fracture in the northern margins of the Arabo-Nubian shield.

Synthesis

Jurassic-Cretaceous and the Neogene movements along the hinge-line of the Coastal Plain are a result of tensional-gravitational fractures of the basement. Movements are associated with facies changes and thickness changes between the two sides of the hinge-line. They are also associated with volcanic activities and erosion which took place mostly in the eastern upthrown block. The longitudinal Coastal Plain fault zone is shown in Fig. 8, Folding movements in the Coastal Plain, as well as over the entire area of Israel took place in one phase during the Senonian to Eocene, and in a rejuvenated phase during the Oligocene (Gvirtzman, 1970). Sediments deposited during folding, namely the Hashephela Group in the first phase, and the Gat Member of the Beit Guvrin Formation as well as the Lakhish Formation in the second phase, do not differ in lithology, environment of deposition and thickness on either side of the hinge-line (Gvirtzman, 1970). It is highly probable that during the Senonian-Oligocene time interval, the Coastal Plain, on both sides of the hinge-line, was under compressional stress. The relief differences between the trough and the eastern uplifted blocks were smoothed by the end of the Turonian, when the folding movement started.

A summary of the tectonic history of the Coastal Plain is shown in Table 4.

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קו-תמורה סטרוקטורלי ודפוזיציונאלי לאורך מישור החוף בישראל - הוכחות לפי מדידות מגנטו-טלוריות

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ת ק צ י ר

קו-תמורה עתיק, המתמשך בכוון צפון-דרום, מצוי בתת-הקרקע של מישור החוף. לאורך קו זה מתמשכים אזור שברים אורכי ושנויים סטרטיגרפיים.

לפי פרופיל מגנטו-טלורי, ניצב לאזור השברים באזור חלץ-אשקלון, יש להסיק כי התשתית הפריקמברית מועתקת מעומק שבין 6.3 ו-8.5 ק"מ בגוש המזרחי המורם, אל עומק של יותר מ-11.6 - 10.6 ק"מ בגוש המערבי המורד. כמו בנגב ובסיני, בהם חסרים סלעים מגיל קמבריום(?) אורדוביק, סילור ודבון, נראה כי גם בגוש המזרחי המורם של מישור החוף, מונחים סדימנטים קרבוניים באי-התאמה, ישירות, על התש-תית. טור הסדימנטים בגוש המזרחי של מישור החוף בנוי מסלעים מגיל קרבון-פרם (שווי זמן של חבורת הנגב) (2 - 1.5 ק"מ), פרם-טריאס (שווי זמן של חבורת רמון) (2 - 1.5 ק"מ), יורה (3 - 2 ק"מ), קרטיקון (בערך ק"מ אחד) ושלישון-רביעון (0.6 ק"מ). ההרכב המשוער של טור הסדימנטים של הגוש המערבי הנו - סלעים אינפרה-קרטיקוניים (יותר מ-8 ק"מ), סלעים מזמן קרטיקון (בערך 1.8 ק"מ) וסלעים מזמן שלישון-רביעון (1.8 ק"מ).

בסלעים מגיל יורה קרטיקון ונאוגן קיימים הבדלים ליתולוגיים, הבדלים בהנאי השקיעה והבדלים בעובי בין שני צידי אזור השברים. השברים מהוים קו-תמורה עתיק שהיה פעיל לפחות מזמן יורה-קרטיקון וכן בזמן הנאוגן. בפרק הזמן שבין הסנון לאוליגוקן, כאשר חלו באזור תנועות קימוט, לא חלה פעילות באזור השברים. הפעילות הטקטונית לאורך אזור השברים קשורה בוולקניזם ובגוש המזרחי המורם גם בארוזיה. לאורך השבר לא חלו תנועות לטרליות.

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מדינת ישראל

משרד הפיתוח
המכון הגיאולוגי

קו תמורה סטרוקטורלי ודפוזיציונאלי לאורך מישור החוף בישראל — הוכחות לפי מדידות מגנטו-טלוריות

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בולטין מס. 55